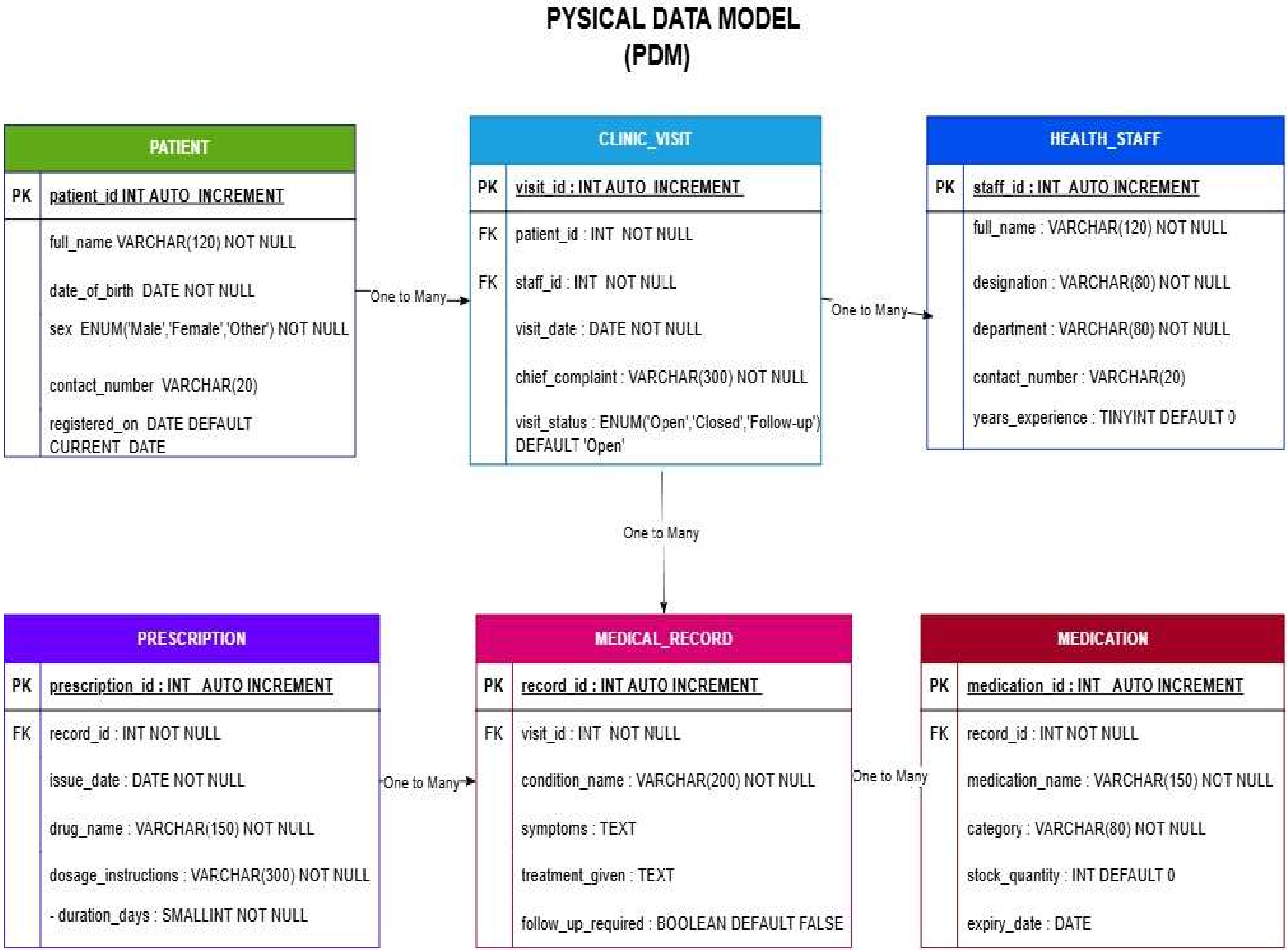
1. INTRODUCTION

The Public Health Clinic Records System is a database solution designed to improve the management of patient information within public health clinics in Sierra Leone. Traditionally, many health facilities rely on paper-based systems, which are inefficient, prone to data loss, and difficult to maintain.

This project aims to develop a relational database using MySQL that stores patient information, clinic visits, medical records, prescriptions, medications, and health staff details. The system enhances data accuracy, improves patient record management, and supports effective healthcare delivery.

The project aligns with Sustainable Development Goal (SDG) 3: Good Health and Well-being by promoting efficient healthcare information management.

REFINED PHYSICAL DATA MODEL (PDM)



The refined Physical Data Model consists of six main entities:

1. Patient
2. Clinic_Visit
3. Health_Staff
4. Medical_Record
5. Prescription
6. Medication

Relationships

- One Patient can register for many Clinic Visits.
- One Health Staff member can handle many Clinic Visits.
- One Clinic Visit can generate many Medical Records.
- One Medical Record can have many Prescriptions.
- One Medical Record can have many Medications.

EXPLANATION OF THE DESIGN DECISIONS

The following design decisions were made during database development:

Primary Keys

Each table contains a unique primary key implemented using AUTO_INCREMENT.

Examples:

- patient_id
- staff_id
- visit_id
- record_id
- prescription_id
- medication_id

Foreign Keys

Foreign keys enforce referential integrity between related tables.

Examples:

- Clinic_Visit.patient_id references Patient.patient_id
- Clinic_Visit.staff_id references Health_Staff.staff_id
- Medical_Record.visit_id references Clinic_Visit.visit_id

Data Types

Different data types were selected based on the nature of the data:

Data Type	Purpose
INT	IDs and numerical values
VARCHAR	Names and short text

DATE	Dates
TEXT	Long descriptions
BOOLEAN	True/False values
ENUM	Limited predefined options

Constraints

The following constraints were used:

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- DEFAULT values
- AUTO_INCREMENT

These constraints improve data consistency and integrity.

DATABASE CREATION SCRIPT

Server: 127.0.0.1

Databases | SQL | Status | User accounts

Run SQL query/queries on server “127.0.0.1”:

```sql
CREATE DATABASE PublicHealthClinicDB;
USE PublicHealthClinicDB;
```

TABLE CREATION SCRIPT

The following tables were created:

```sql
USE PublicHealthClinicDB;
CREATE TABLE Patient (
    patient_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(120) NOT NULL,
    date_of_birth DATE NOT NULL,
    sex ENUM('Male','Female','Other') NOT NULL,
    contact_number VARCHAR(20),
    home_address VARCHAR(250),
    registered_on DATE DEFAULT CURRENT_DATE
);
```

Patient

Health_Staf

```sql
USE PublicHealthClinicDB;
CREATE TABLE Health_Staff (
    staff_id INT AUTO_INCREMENT PRIMARY KEY,
    full_name VARCHAR(120) NOT NULL,
    designation VARCHAR(80) NOT NULL,
    department VARCHAR(80) NOT NULL,
    contact_number VARCHAR(20),
    years_experience TINYINT DEFAULT 0,
    is_active BOOLEAN DEFAULT TRUE
);
```

Run SQL query/queries on database publichealthclinicdb:

```sql
INSERT INTO Clinic_Visit
(patient_id, staff_id, visit_date, chief_complaint)
VALUES
(1,4,'2026-06-01','Fever and headache'),
(2,5,'2026-06-02','Persistent cough'),
(3,6,'2026-06-03','Stomach pain');
```

Clinic_Visit

Medical_Record

Prescription

Medication

Run SQL query/queries on table publichealthclinicdb.clinic_visit:

Run SQL query/queries on table publichealthclinicdb.medical_record:

```sql
INSERT INTO Prescription
(record_id, issue_date, drug_name, dosage_instructions, duration_days)
VALUES
(7,'2026-06-01','Coartem','2 tablets twice daily',3),
(8,'2026-06-02','Paracetamol','1 tablet every 8 hours',5),
(9,'2026-06-03','Omeprazole','1 capsule daily',14);
```

```sql
USE PublicHealthClinicDB;
CREATE TABLE Medication (
    medication_id INT AUTO_INCREMENT PRIMARY KEY,
    record_id INT NOT NULL,
    medication_name VARCHAR(150) NOT NULL,
    category VARCHAR(80) NOT NULL,
    stock_quantity INT DEFAULT 0,
    expiry_date DATE,

    FOREIGN KEY (record_id)
    REFERENCES Medical_Record(record_id)
);
```

SAMPLE DATA INSERTION

Sample data was inserted into all tables to test database functionality.

```sql
INSERT INTO Patient
(patient_id,full_name, date_of_birth, sex, contact_number, home_address)
VALUES
(1,'Mariama Kamara','2000-05-15','Female','076123456','Freetown'),
(2,'Abdul Sesay','1998-03-10','Male','077234567','Bo'),
(3,'Fatmata Bangura','2002-11-21','Female','078345678','Kenema');
```

Patients

Staff

```sql
INSERT INTO Health_Staff
( staff_id,full_name, designation, department, contact_number, years_experience)
VALUES
(4,'Dr. John Kanu','Doctor','General Medicine','075111111',10),
(5,'Nurse Hawa Conteh','Nurse','Outpatient','075222222',6),
(6,'Dr. Ibrahim Koroma','Doctor','Pediatrics','075333333',8);
```

Run SQL query/queries on table publichealthclinicdb.clinic_visit:

```sql
 INSERT INTO Medical_Record
(visit_id, condition_name, symptoms, treatment_given, follow_up_required)
VALUES
(1,'Malaria','Fever, headache','Antimalarial drugs',TRUE),
(2,'Flu','Cough, cold','Flu medication',FALSE),
(3,'Gastritis','Abdominal pain','Antacid treatment',TRUE);
```

Medical Records

Prescriptions

Run SQL query/queries on table publichealthclinicdb.medical_record:

```sql
INSERT INTO Prescription
(record_id, issue_date, drug_name, dosage_instructions, duration_days)
VALUES
(7,'2026-06-01','Coartem','2 tablets twice daily',3),
(8,'2026-06-02','Paracetamol','1 tablet every 8 hours',5),
(9,'2026-06-03','Omeprazole','1 capsule daily',14);
```

Run SQL query/queries on table publichealthclinicdb.medical_record:

```sql
INSERT INTO Medication
(record_id, medication_name, category, stock_quantity, expiry_date)
VALUES
(7,'Coartem','Antimalarial',50,'2027-01-01'),
(8,'Paracetamol','Pain Relief',100,'2027-05-01'),
(9,'Omeprazole','Gastrointestinal',70,'2027-03-01');
```

Medication

UPDATE AND DELETE DEMONSTRATION

UPDATE Statement

The patient's contact number was updated.

Run SQL query/queries on table publichealthclinicdb.medical_record:

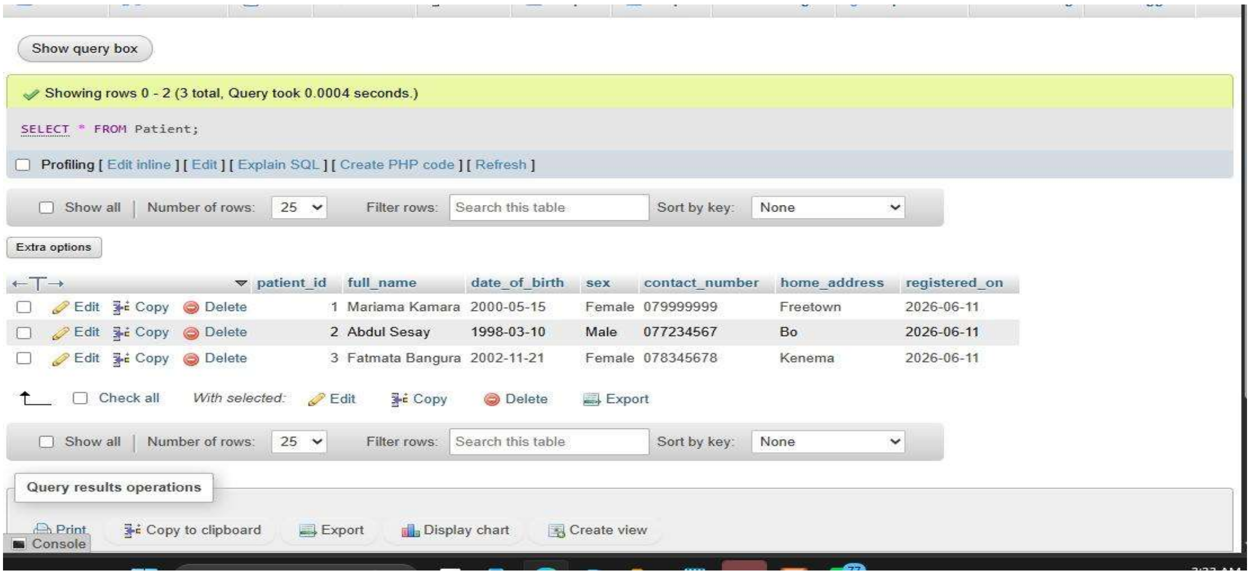
```sql
UPDATE Patient
SET contact_number = '079999999'
WHERE patient_id = 1;
```

DELETE Statement

SQL QUERIES AND RESULTS

Query 1: Display All Patients

Result:

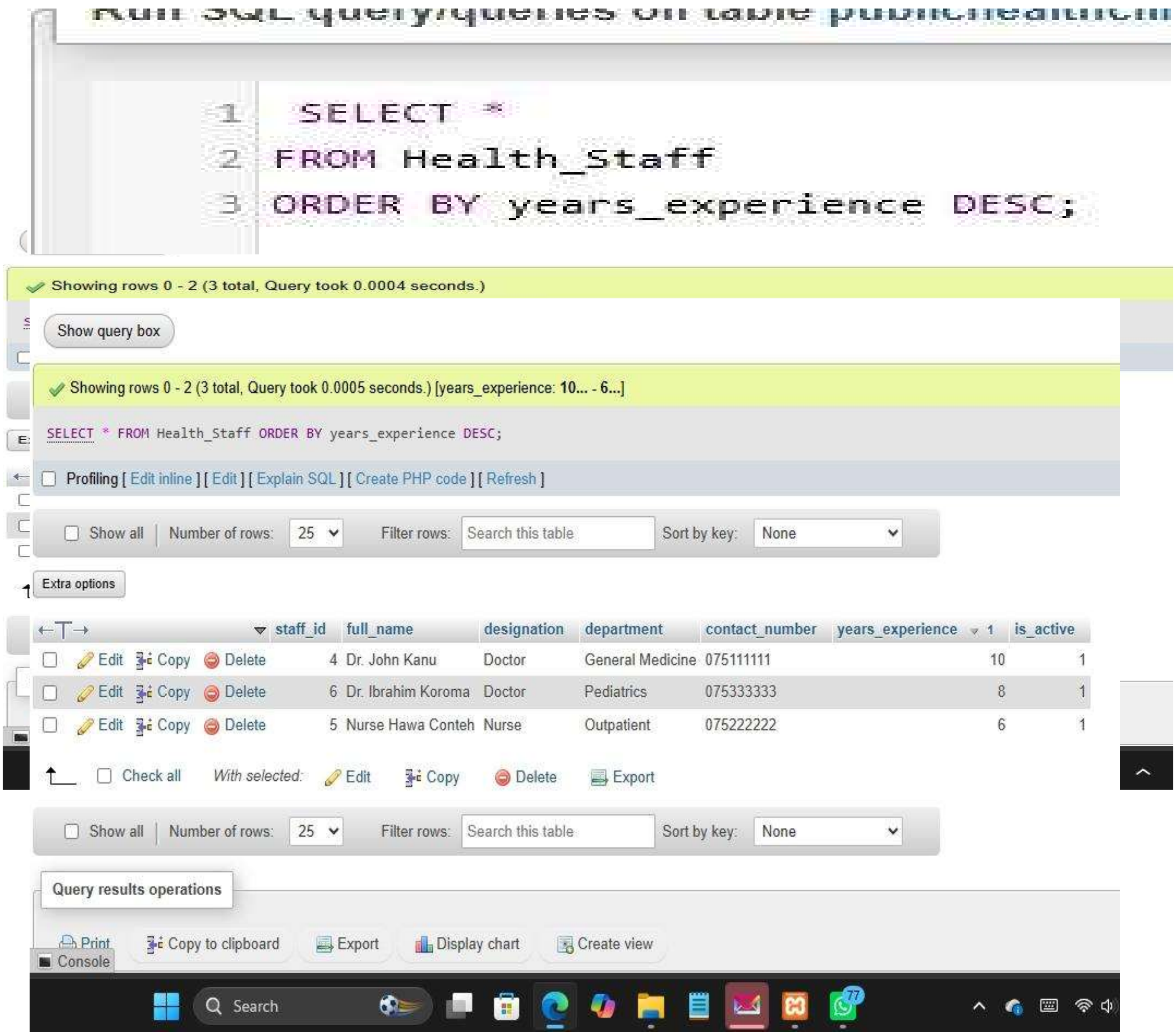


	patient_id	full_name	date_of_birth	sex	contact_number	home_address	registered_on
Edit Copy Delete	1	Mariama Kamara	2000-05-15	Female	079999999	Freetown	2026-06-11
Edit Copy Delete	2	Abdul Sesay	1998-03-10	Male	077234567	Bo	2026-06-11
Edit Copy Delete	3	Fatmata Bangura	2002-11-21	Female	078345678	Kenema	2026-06-11

Query 2: Open Clinic Visits

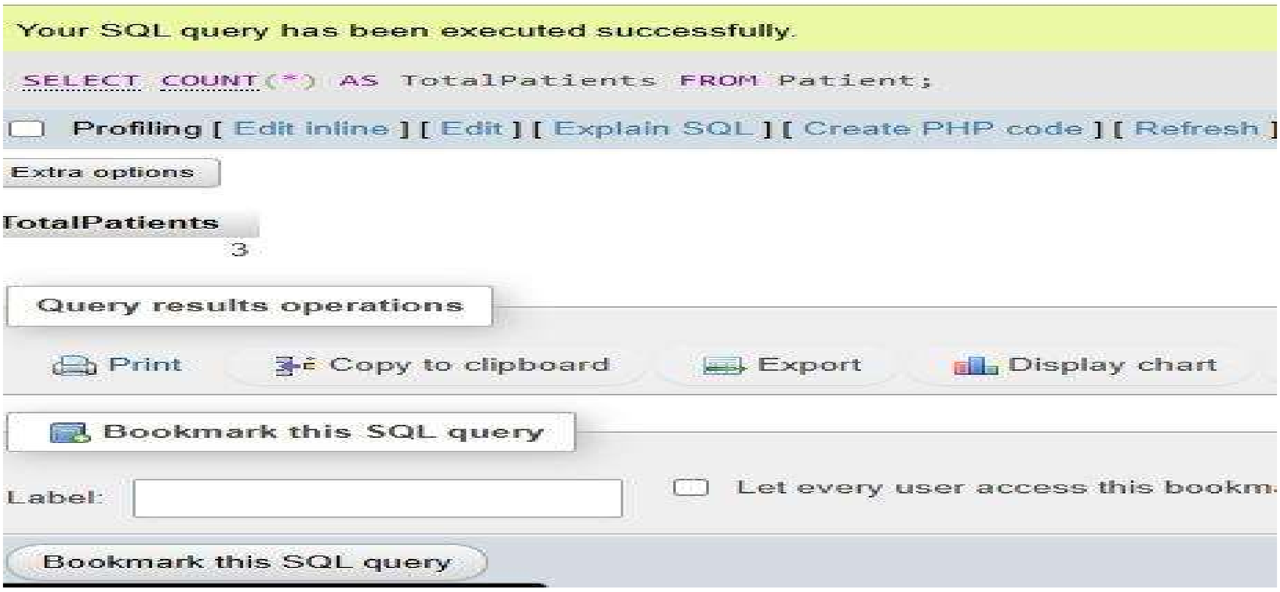
Result:

Query 3: Sort Staff by Experience



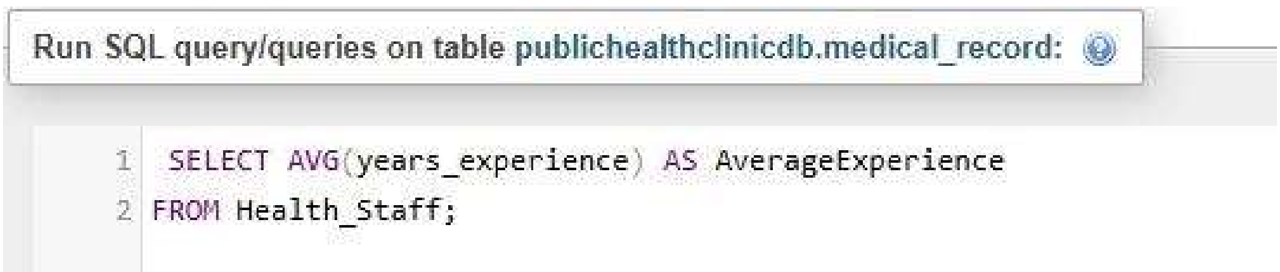
Result:

Query 4: Count Total Patients



Result:

Query 5: Average Staff Experience



Result:

USER MANAGEMENT

Database security is important to prevent unauthorized access.

Create User

CREATE USER 'groupmember1'@'localhost'

IDENTIFIED BY 'Password123';

Change Password

ALTER USER 'groupmember1'@'localhost'

IDENTIFIED BY 'NewPassword123';

Grant Privileges

GRANT ALL PRIVILEGES

ON PublicHealthClinicDB.*

TO 'groupmember1'@'localhost';

FLUSH PRIVILEGES;

Importance of Database Security

- Protects patient information.

- Prevents unauthorized access.
- Maintains confidentiality.
- Preserves data integrity.
- Ensures accountability through user permissions.

CONCLUSION

The Public Health Clinic Records System was successfully designed and implemented using MySQL. The project demonstrated the practical application of database concepts including database creation, table implementation, relationships, constraints, data manipulation, querying, and user management.

The system provides an efficient method for managing healthcare records and contributes to improved healthcare service delivery in Sierra Leone. Through this project, valuable skills in relational database design and implementation were developed and applied successfully.